Performance Testing

Food Ordering Behaviour and Consumer Trends

Objective

Performance testing was conducted to ensure the Tableau dashboard and story operate efficiently, load quickly, and respond accurately to user interactions once embedded in the Flask web application. The testing verifies that all visualizations, filters, KPIs, and calculated fields function as expected.

Amount of Data Loaded

The dashboard was tested against the complete food ordering dataset, covering every Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, and Time Taken for Delivery record, to confirm that Tableau renders the full dataset without lag or truncation.

Utilization of Data Filters

Filter	Purpose
City	Isolate order and delivery performance by location
Cuisine	Compare demand and revenue across cuisine types
Meal Type	Analyze breakfast / lunch / dinner / snack demand

Each filter was tested individually and in combination to confirm that all connected charts and KPI cards update correctly and instantly.

No. of Calculation Fields

Calculated Field	Purpose
Total Orders	Counts total number of orders (Order ID)
Total Order Value	Sums Order Value across all orders
Average Delivery Time	Averages Time Taken for Delivery
Average Order Value	Order Value divided by Total Orders
Delivery Fee % of Order Value	Delivery Fee divided by Order Value
Top Cuisine	Identifies the cuisine with the highest order count

No. of Visualizations / Graphs

#	Visualization	Chart Type
1	Total Orders	KPI Card
2	Total Order Value	KPI Card

3	Average Delivery Time	KPI Card
4	Cuisine Analysis	Bar Chart
5	Meal Type Distribution	Pie / Donut Chart
6	City-wise Demand	Map / Bar Chart
7	Delivery Time Distribution	Histogram / Box Plot
8	Order Value vs Delivery Fee	Scatter Plot

Test Cases

Test Case 1: Dashboard Loading

Objective: Verify the dashboard loads successfully. | Expected Result: Dashboard opens without errors and displays all visualizations. | Status: Passed

Test Case 2: KPI Validation

Objective: Verify all KPI values are calculated correctly. | Expected Result: Total Orders, Total Order Value, and Average Delivery Time display accurate values. | Status: Passed

Test Case 3: Filter Functionality

Objective: Verify dashboard filters work correctly. | Expected Result: Selecting City, Cuisine, or Meal Type updates all related charts and KPIs dynamically. | Status: Passed

Test Case 4: Chart Accuracy

Objective: Verify all charts display correct information. | Expected Result: Bar, pie, map, and scatter charts accurately represent the dataset. | Status: Passed

Test Case 5: Story Navigation

Objective: Verify navigation between Tableau Story scenes. | Expected Result: Users can move between scenes smoothly without errors. | Status: Passed

Test Case 6: Web Embed Responsiveness

Objective: Verify the dashboard and story render correctly inside the Flask app across screen sizes. | Expected Result: The tableau-viz component resizes and displays correctly on desktop and mobile widths. | Status: Passed

Performance Results

- Dashboard and story load successfully from Tableau Public.
- KPIs and calculated fields are computed accurately.
- Interactive filters respond correctly and instantly.
- Visualizations update without noticeable delay.
- Dashboard layout remains organized across different screen sizes.
- Story navigation functions correctly scene to scene.

Conclusion

The performance testing confirms that the Food Ordering Behaviour and Consumer Trends dashboard and story are reliable, accurate, and responsive. All filters, calculated fields, and visualizations perform as expected, both in Tableau and when embedded in the Flask web application.